

MONSTER ORBIFOLD E_3 -TOPOLOGICAL DESCENT CRITERION VIA THE LEECH $\mathbb{Z}/2$ -ORBIFOLD

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ABSTRACT. The Moonshine module $V^\natural$ enters through a finite-orbifold boundary construction. The E_3 -topological object is the descended holomorphic-topological factorisation algebra obtained from the Leech lattice model; its boundary chiral algebra is $V^\natural$. A single-colour E_3 -topological algebra on the original boundary VOA complex requires additional data. The descent requires the orbifold BV anomaly to vanish and the FLM twisted-sector construction to identify the boundary VOA. The van Ekeren–Möller–Scheithauer classification supplies cyclic orbifold presentations for the Schellekens list; anomaly vanishing and BV descent remain type-specific inputs.

1. THE LANDSCAPE AFTER THE LEECH $\mathbb{Z}/2$ CONSTRUCTION

Class	E_3 -topological scope	Mechanism
Affine KM at $k \neq -h^\vee$	original complex	Costello–Gaiotto
Class- $\mathcal{M}$ DS algebras	weight-completed; strict original chain lift conditional	DS–Hochschild
Free PVAs	Conditional original-complex lift	Khan–Zeng anomaly
Heisenberg / lattice	original complex	Abelian hCS
$V^\natural$ (Moonshine)	Descended HT bulk with boundary $V^\natural$	Leech $\mathbb{Z}/2$ orbifold
71 Schellekens $c = 24$ VOAs	Type-specific finite-orbifold descent	EMS cyclic pres
Class- $\mathcal{M}$ principal (Vir , $\mathcal{W}_N$)	weight-completed	Half-BRST + M
Non-principal non-orbifold original-complex lift	no general theorem	none known

Finite-orbifold descent gives an original-complex E_3 model only after two separate inputs have been supplied: anomaly-free BV descent for the finite group action and identification of the boundary VOA. The Leech $\mathbb{Z}/2$ construction is therefore an exceptional lattice route, not a generic class- $\mathcal{M}$ theorem. Generic class- $\mathcal{M}$ principal algebras outside this finite-orbifold reach remain governed by the weight-completed/pro-ambient theorem and the conditional strict E_3 chain-lift statement.

2. STRUCTURAL SCOPE OF THE ORBIFOLD THEOREM

The Monster descent statement separates the VOA theorem from the HT descent theorem. Frenkel–Lepowsky–Meurman construct the boundary VOA. Dijkgraaf–Witten theory places finite-group gauging obstructions in $H^3(BG; \mathbb{U}(1))$. The HT conclusion follows only after the Leech $\mathbb{Z}/2$ BV anomaly has been shown to vanish.

Theorem 2.1 (Monster orbifold E_3 -topological descent criterion). *Let $V^\natural = V_\Lambda^+ \oplus V_\Lambda^{\text{tw},+}$ be the FLM Moonshine module (the $\tilde{\sigma}$ -invariant part of the $\mathbb{Z}/2$ -orbifold of the Leech lattice VOA, with the standard FLM lift, canonical up to VOA isomorphism). Suppose:*

- (i) *the Leech lattice HT model $\mathcal{F}_\Lambda$ carries the abelian chain-level E_3 -topological structure;*

- (ii) the lifted $\mathbb{Z}/2$ action has trivial finite-orbifold BV anomaly $[\alpha_{\text{orb}}] = 0$ in $H^3(B\mathbb{Z}/2; \mathbb{U}(1))$;
- (iii) finite-orbifold BV descent applies to $\mathcal{F}_\Lambda$.

Then the descended orbifold factorisation algebra $\mathcal{F}_\Lambda^{\text{orb}}$ carries a chain-level E_3 -topological algebra structure, and its boundary chiral algebra is $V^\natural$.

Proof. FLM boundary algebra. The FLM twisted module V_Λ^{tw} has half-spinor ground-state multiplicity 2^{12} , and FLM Chapters 8–10 prove the twisted Jacobi identity at chain level. Thus $V_\Lambda^+ \oplus V_\Lambda^{\text{tw},+}$ is the Moonshine module $V^\natural$.

Finite-group anomaly. For a finite group G , the obstruction to gauging a holomorphic orbifold sector is a Dijkgraaf–Witten class in $H^3(BG; \mathbb{U}(1))$. For $G = \mathbb{Z}/2$ this group is $\mathbb{Z}/2$; in a normalised representative the class is detected by the sign $\alpha(\sigma, \sigma, \sigma) = \pm 1$. A normalized 2-coboundary has value 1 on (σ, σ, σ) , so this sign is a class invariant. Hypothesis (ii) is precisely the assertion that the Leech lift gives the sign +1.

Descent. Hypothesis (i) supplies the E_3 -topological operations before gauging. Hypotheses (ii)–(iii) say that the finite orbifold BV construction has no anomalous associator and therefore inherits those operations on the descended factorisation algebra. Boundary restriction commutes with the invariant and twisted-sector decomposition, so the boundary chiral algebra of $\mathcal{F}_\Lambda^{\text{orb}}$ is the FLM orbifold VOA $V^\natural$. $\square$

3. SCHELLEKENS 71 AND THE EMS BIJECTION

The cyclic-orbifold literature supplies VOA presentations, not an automatic HT descent theorem. Van Ekeren–Möller–Scheithauer develop cyclic orbifold theory for holomorphic VOAs; van Ekeren–Lam–Möller–Shimakura show that every strongly rational holomorphic $c = 24$ VOA with $V_1 \neq 0$ is obtained from the Leech lattice VOA by a cyclic orbifold; Höhn–Möller give systematic constructions from Niemeier lattice VOAs. The HT statement still needs the anomaly and BV descent hypotheses.

- (a) *Type A* (pure Niemeier, 24 cases): no finite orbifold is being gauged; the lattice HT model is the abelian case.
- (b) *Type B* (Leech $\mathbb{Z}/2$, 1 case): the Moonshine module $V^\natural$; Theorem 2.1.
- (c) *Type C* (non-trivial cyclic orbifolds): level matching and the deep-hole tables identify the VOA presentation; the Dijkgraaf–Witten class and BV descent must still be verified for the chosen cyclic action.

Theorem 3.1 (Schellekens cyclic-orbifold descent criterion). *For a Schellekens $c = 24$ VOA presented as a cyclic orbifold of a Niemeier lattice VOA, if the cyclic level-matching condition holds, the Dijkgraaf–Witten anomaly class vanishes, and finite-orbifold BV descent applies to the chosen lattice HT model, then the descended HT factorisation algebra carries an E_3 -topological structure whose boundary chiral algebra is the corresponding Schellekens VOA.*

The full 71-case statement is therefore a criterion until the type-specific anomaly calculations are supplied: Type A is vacuous, Type B is the Leech $\mathbb{Z}/2$ calculation, and Type C is a cyclic level-matching plus anomaly computation. The EMS classification itself supplies cyclic presentations, not the BV descent theorem.

4. POSITION IN THE TOPOLOGIZATION LANDSCAPE

The landscape separates five mechanisms:

- affine Kac–Moody $V_k(\mathfrak{g})$ at $k \neq -h^\vee$ (Costello–Gaiotto Sugawara route),

- the class- $\mathcal{M}$ DS landscape in the weight-completed/pro ambient, with strict original-complex chain lift only in the stated conditional scope,
- freely generated PVAs, conditional on the Khan–Zeng analytic SDR input,
- $V^\natural$ and the Schellekens cyclic-orbifold cases when the type-specific anomaly and BV-descent inputs hold,
- Heisenberg and all lattice VOAs (abelian hCS).

Chain-level E_3 -topological structure on the weight-completed bar complex is proved for class- $\mathcal{M}$ principal algebras (Virasoro, $\mathcal{W}_N$) via the DS–Hochschild and MC₅ pattern; the strict original-complex chain lift has the conditional status recorded in the chiral Higher Deligne and universal-holography chapters. The Leech $\mathbb{Z}/2$ construction does *not* change that generic class- $\mathcal{M}$ scope, because $V^\natural$ is reached by finite-orbifold descent from the class-G lattice model.

5. PRIMARY SOURCES AND LOCAL HYPOTHESES

The source inputs are:

- Frenkel–Lepowsky–Meurman construct $V^\natural$ as the Leech $\mathbb{Z}/2$ orbifold and prove the twisted Jacobi identity.
- Dijkgraaf–Witten identify finite-group orbifold anomalies with group-cohomology classes; Dijkgraaf–Pasquier–Roche identify the same class as the associator twist in the finite-group orbifold tensor category. This note uses only the class $[\alpha] \in H^3(BG; \mathbb{U}(1))$, not a Drinfeld-double equivalence as an E_3 -descent theorem.
- Dong–Li–Mason prove modular invariance of orbifold trace functions under their finiteness hypotheses. Modular invariance is evidence for level matching; it is not by itself the BV descent theorem.
- Van Ekeren–Möller–Scheithauer, van Ekeren–Lam–Möller–Shimakura, and Höhn–Möller supply cyclic-orbifold constructions and Schellekens-list control for holomorphic $c = 24$ VOAs.
- The local HT inputs are the abelian Leech lattice E_3 -topological model, the explicit vanishing of the relevant finite-orbifold BV anomaly, and the finite-orbifold BV descent theorem. If any of these three inputs is absent, the conclusion above remains conditional.